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2pSC17. Augmented visual feedback in second language learning: Training Japanese post-alveolar flaps to American English speakers

June S. Levitt and William F. Katz

The purpose of this study is to better understand the role of augmented visual feedback during speech motor learning in healthy individuals. An experiment was conducted to examine how adult monolingual speakers of American English learn to produce a non-native consonant: the Japanese post-alveolar flap. Eight participants were randomly assigned to one of two training groups: (1) conventional L2 training and (2) conventional L2 training with Electromagnetic Articulography (EMA)-based kinematic feedback on tongue tip position. Participants completed three baseline recording sessions, eight training sessions, and two post-training recording sessions. During training, participants practiced a total of 16 disyllabic non-words containing the post-alveolar flap. Three repetitions of the trained items were recorded as probes at each session. Duration was analyzed as an acoustic marker of flap quality, as this feature is known to play a role in the perception of this consonant. The current report must be considered tentative due to unblinded test conditions. Nevertheless, the results suggest (1) before training, participants produced substantially longer flap durations than native Japanese speakers; (2) with training, flap durations became closer to those of native norms; and (3) kinematic feedback with EMA appeared to facilitate the acquisition and maintenance of novel speech sound production.

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Augmented visual feedback in second language learning: Training Japanese post-alveolar flaps to American English speakers

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INTRODUCTION

Researchers agree that feedback plays an important role in the process of acquiring various motor behaviors (Schmidt & Lee, 1999; Shea & Wulf, 2005). Feedback provides information on the current status of the learner's movements by allowing the learner to associate improved motor behavior with sensory information. Feedback information may be divided into two categories, intrinsic and extrinsic (Schmidt & Wrisberg, 2004). Intrinsic feedback is the sensory information that is obtained from a learner's perception of a task and from his/her response to it without special assistance from other sources. Extrinsic feedback (also referred to as "augmented feedback") is provided to the learner by outside sources, such as comments from instructors or the digital display of a stopwatch (Schmidt & Wrisberg, 2004).

Kinematic feedback is an important type of augmented feedback involving information about movement that the learner cannot directly perceive (Schmidt & Wrisberg, 2004). The term "kinematic" refers to measures of pure motion. Feedback about kinematics could relate to the position, time, velocity, and patterns of the coordination of the movements. Kinematic feedback may be provided by people with comments about movements such as "You bent your elbow that time," or by instruments (such as systems that display movements on a computer screen).

Studies have suggested positive effects of kinematic feedback in various aspects of motor learning. The majority of these studies on kinematic feedback have focused on limb motor control, not speech. One reason for this preponderance of limb motor studies is that external, fixed articulators such as the arm are relatively easy to study, whereas the less visible speech articulators (tongue and velum) are not. Also, the movement patterns of the tongue, a muscular hydrostat, are arguably quite different from those of jointed limbs (Kent, 2000). However, recent studies have begun to draw interesting parallels between limb and speech motor control (Ballard, Granier, & Robin, 2000; De Nil & Lafaille, 2002; Dowd, Smith, & Wolfe, 1998; Hageman, Robin, Moon, & Folkins, 1994; Hardison, 2003; Katz, Bharadwaj, & Carstens, 1999; Katz, Bharadwaj, Gabbert, & Stettler, 2002). These studies suggest that motor training principles derived from the limb motor literature can be successfully applied to speech.

In one important study, De Nil and Lafaille (2002) examined whether the presence of visual kinematic feedback differently influences the accuracy of finger and jaw movements. Participants were instructed to precisely move either the index finger on the dominant (right) hand, or the jaw, to a predefined target position, while online visual feedback of the moving articulator was either present or removed. Improved accuracy was reported with visual feedback, regardless of whether the movements were performed with the finger or the jaw. Although this study did not involve speech production, the results indicated potential benefit of kinematic feedback in one of the speech articulators.

Dowd and colleagues investigated the effect of visual presentation of acoustic measurements of the vocal tract on speech production training (Dowd, Smith, & Wolfe, 1998). Healthy adult speakers of English learned to produce French vowels either (1) with online visual feedback of the first two resonances of the vocal tract, or (2) with only traditional auditory feedback. The authors hypothesized that feedback that does not involve auditory information may be more effective for adult L2 learners, because they will not be influenced by the categorical perception based on the speech sounds in their native languages. Online feedback of the resonances of the vocal tract (R1 and R2) is known to correspond to the vertical and horizontal positions of the tongue (Stevens, 2000). Superior acoustic

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properties and recognizability of the vowels were reported when the participants used the visual display of vocal-tract acoustics, thereby, demonstrating a positive effect of visual kinematic feedback for speech production training.

Whereas the technique used by Dowd and colleagues required experimental participants to infer tongue movement from visual feedback (derived from acoustics), there have been other experiments that have used techniques to directly transduce speech movements as the basis for feedback. These technologies include Electropalatography (EPG) and Electromagnetic Articulography (EMA). EPG shows changing tongue-palate contact patterns in real time during speech, using an apparatus that fits onto the upper surface of the mouth and contains built-in electrodes. EMA, the instrument used in the present study, provides a non-invasive method of tracking speech articulator movements in the midsagittal plane. This technique employs low-field strength, alternating electromagnetic fields to track small sensors attached to the articulators (Schönle, Grabe, Wenig, Höhne, Schrader, & Conrad, 1987). EMA can provide real-time visual representation of tongue movements.

In one application, Katz and colleagues used EMA-based visual kinematic feedback to treat speech production errors of individuals with Apraxia of Speech (AOS) (Katz et al., 1999). To assess the effectiveness of experimental programs for the treatment of AOS, Katz et al. (1999) provided two different treatment conditions to an individual with AOS. These conditions were (1) a visually guided biofeedback treatment with EMA concerning tongue-tip position and (2) a foil treatment in which a computer program delivered voicing-contrast stimuli for simple repetition. They reported rapid acquisition and maintenance of articulatory movements in the EMA experimental condition, whereas the control condition produced short-term improvement but poor maintenance. Thus, the results suggest positive effects of EMA augmented visual feedback on speech motor learning for individuals with AOS.

Most limb motor learning studies have tested healthy individuals. However, several recent studies have extended the work on principles of motor learning to patients with neurological disease in re-learning limb control (Goodgold-Edwards & Cermak, 1990; Hanlon, 1996; Jarus, 1994; Sabari, 1991; Stevans & Hall, 1998). In general, motor learning principles appear to apply similarly in the intact and the neurologically impaired system (Winstein, Wing, & Whitall, 2003). It is reasonable then to hypothesize that the same principles which enhance limb motor learning will also apply to speech motor learning in both healthy and neurologically impaired individuals (Kim, 2007).

The goal of this study is to examine the role of kinematic feedback of the tongue during novel speech learning in healthy individuals using EMA. Specifically, the present study investigates the effects of EMA-based augmented visual feedback for healthy monolingual adult speakers of American English learning a non-native consonant: the Japanese post-alveolar flap. These data will provide important background for interpreting the results of EMA intervention studies with disordered individuals (e.g., Katz et al., 1999, 2002).

The main research question concerns the relationship between limb and speech motor learning with augmented visual feedback. It is predicted that augmented visual feedback of tongue tip position provided by EMA will facilitate speech acquisition, as described for limb motor behavior (De Nil & Lafaille, 2002). The retention of learned movement after the removal of visual feedback will also be explored. Although frequent augmented feedback may guide participants during the training, some researchers have described a detrimental effect of high-frequency feedback when measured by a retention test in the absence of such feedback (Salmoni, Schmidt, & Walter, 1984; Schmidt, 1991, 1997; Winstein & Schmidt, 1990). These researchers suggest that the learner may rely on concurrent feedback as a type of "crutch" to produce the correct action during practice. Excessive reliance on feedback prevents the learner from exercising his/her own capabilities necessary for the retention test, when augmented feedback is removed. Thus, performance deteriorates when the feedback is not present.

In contrast, other researchers have reported that the effect of concurrent feedback may extend to the retention test phase (Kinkade, 1963; Swinnen, Lee, Verschueren, Serrien, & Bogaerds, 1997; De Nil & Lafaille, 2002). These researchers suggested that the augmented feedback effectively strengthens intrinsic feedback and thereby facilitates motor behavior.

Different explanations have been offered to reconcile these opposing views concerning the role of feedback. These explanations involve the degree to which feedback matches the trained behavior and whether participants can successfully link the augmented feedback to intrinsic feedback throughout training (Swinnen, 1996). It is far from clear how these factors relate to speech motor behavior, making it difficult to predict *a priori* how extrinsic kinematic feedback will affect speech. However, based on positive reports of EMA treatment of disordered speech (Katz et al., 1999, 2002), it is predicted trained behaviors will also be retained in post-training probes.

A secondary series of issues address the predictions of Schema theory (Schmidt, 1975), an influential view concerning the planning and execution of motor behavior. Schema theory proposes that learners construct Generalized Motor Programs (GMP) instead of learning specific movements. A GMP is a set of motor instructions that defines a pattern or class of movement (Schmidt & Wrisberg, 2004). According to Schema Theory, different movements within a GMP are performed by varying parameters to meet altered environmental demands. For example, the general action of writing one's name would be coded by a GMP (or set of GMPs). Whether writing on

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a piece of paper or on a blackboard would involve different sets of parameters, such as different speeds, displacements, or forces.

Despite much discussion concerning the general concept of a GMP, it has proven difficult to determine exactly what a GMP is in speech. Some researchers have suggested that potential GMP organization centers on the individual phoneme (Knock, Ballard, Robin, & Schmidt, 2000), other featural properties such as manner (Van der Merwe, 1997) or place of articulation (Katz et al., 1999, 2002). In the present case, we shall make the parsimonious assumption that learners form one GMP for the phoneme /ɽ/ that will generalize across vowel conditions.

In the present experiment, post-consonantal vowel context was manipulated for two reasons. First, Japanese phonology relies heavily on disyllables and there has been some discussion in the Japanese literature suggesting that certain Consonant-Vowel (CV) combinations involve different types of flap articulation than others (Kobayashi, 2005; but see Okada, 1993). Second, by manipulating post-consonantal context, it may be possible to gain insights about the nature of the GMP. If participants are trained on one set of CV combinations and learning is found to generalize to other CV combinations, this would suggest a unitary GMP for the consonant phoneme.

An additional factor manipulated is the position of the flap phoneme in the syllable. In American English, alveolar stops (/t, d/) become a voiced alveolar flap between a stressed vowel and unstressed vowel (e.g., "*Betty*", "*ladder*"). That is, American English flap only occurs in word-medial position. However, Japanese flaps are found both in word-initial and medial positions. Therefore, word-initial flaps will be less familiar for native speakers of American English. According to one theory, learning processes are positively influenced by familiarity (Davidson, 2005, 2006). On this view, word-initial Japanese flaps (less familiar) should be more difficult for American talkers than for these flaps in medial position. In contrast, Flege's Speech Learning model (SLM) suggests that perception plays an important role and talkers tend to assimilate L2 sounds to familiar sounds in their native language (Flege, 1995). According to this theory, less familiar sounds are more learnable. As such, this theory would predict that word-medial (more familiar) Japanese flaps would be more difficult for native speakers of American English. These theories are investigated in the current experiment by including four different disyllables /ɽɑ/, /ɽɛ/, /ɽi/, and /ɽo/ as stimuli.

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METHOD

Participants

Eight healthy adults participated in the training sessions. The participants were monolingual female native speakers of American English with a minimum of 12 years of formal education (mean age=28, SD=9.35, range=22-48) and no reported history of neurological, language, learning, speech, or hearing deficits.

Since previous studies have demonstrated that (1) speech/language acquisition changes through childhood and adolescence, and (2) aging alters the functional processing of speech cues, the present research used participants between 18 and 60 years of age (Cienkowski & Carney, 2002; Tremblay, Piskosz, & Souza, 2002).

Although gender differences in learning foreign language have been reported in terms of vocabulary building (Catálan, 2003; Grace, 2000; Green & Oxford, 1995), there are no reports suggesting sex-specific different outcomes with regard to L2 (second language) speech sound acquisition. The current investigation used only female participants to eliminate potential confounds due to male/female vocal tract differences (see Bachorowski & Owren, 1999; Kent & Read, 2002; Klatt & Klatt, 1990; Titze, 1989). Female voices are approximately one octave higher than male's. This difference in fundamental frequency relates primarily to the shorter vocal tract length of female than male (Titze, 1989). Also, female voices tend to have a greater excitation on F3 than male voices (Klatt & Klatt, 1990) and greater interaction between source and filter (Kent & Read, 2002).

To exclude individuals who have difficulties perceiving novel speech sounds, all participants were required to pass a brief screening task prior to the speech training experiment. This was a basic same-different (AX) task with CV-syllables, in which two of the three pre-recorded syllables (American English /dɑ/, /lɑ/, and Japanese flap /ɽɑ/) were presented as pairs, via the Direct RT Research software program. The pre-recorded sounds were produced by a single talker (JL), amplitude balanced, and acoustically matched in intonation contour and formant frequencies over their vowel portions. Participants adjusted the intensity of the stimuli to a comfortable listening level. All six possible combinations of the three sounds (three "same" and three "different" combinations) were repeated five times, for a total of 30 pairs. Following Logan (1993), participants were required to perform 80% or higher on this task to qualify to participate in the training study. All participants passed this screening procedure, with score averaging 96% correct discrimination (SD=5.8%, range=83%-100%).

Experiment: Japanese flap production

Stimuli

A total of 16 Japanese words that included the post-alveolar flap /ɽ/ were used in this experiment. The sound was selected for training because (1) it is not familiar to native speakers of English, and (2) its kinematic properties can be well described by tracking tongue movement in the mid-sagittal plane with an EMA system. All stimuli were pronounceable as disyllabic utterances containing /ɽ/ at the word-initial or word-medial position, with different post-consonantal vowel contexts /ɑ/, /ɛ/, /i/, or /o/ (see Table 1). All words have primary stress on the first syllable. Although some of these words constitute meaningful words in Japanese, they are effectively used as non-words for this foreign speech sound training study.

Japanese /ɽ/ is generally described as a post-alveolar flap consonant (Koizumi, 2003; Okada, 1993), articulated with a similar manner as the English flap /ɾ/. However, the Japanese flap is characterized by a shorter duration and a more posterior place of articulation than the English flap (Koizumi, 2003; Okada, 1993). A pilot study for the present research indicated that the average duration of Japanese /ɽ/ in disyllabic words produced by four native speakers of Japanese was 9.5% of the total utterance, and that of four native speakers of American English was 17.1%. In absolute duration terms, this is 33 ms and 56 ms, respectively. Previous work examining temporal properties of stops vs. flaps have indicated that differences of this magnitude provide strong cues for this manner distinction (Kent & Read, 2002). Based on these data, flap duration was chosen as a measure of sound production quality.

TABLE 1. Stimuli

Set A		**Set B**	
/ɽɑ/ sequence	**/ɽɛ/ sequence**	**/ɽi/ sequence**	**/ɽo/ sequence**
/ˈhɑɽɑ/	/ˈhɛɽɛ/	/ˈhiɽi/	/ˈhoɽo/
/ˈtoɽɑ/	/ˈtoɽɛ/	/ˈtoɽi/	/ˈtoɽo/
/ˈɽɑkɯ/	/ˈɽɛkɯ/	/ˈɽikɯ/	/ˈɽokɯ/
/ˈɽɑtɑ/	/ˈɽɛtɑ/	/ˈɽitɑ/	/ˈɽotɑ/

Procedure

Participants completed eight training sessions preceded by three baseline recording sessions and followed by two post-training recording sessions. The post-training sessions were conducted four weeks after the last training session. Training sessions were conducted 2x per week in the Speech Production Lab at the University of Texas at Dallas. Participants were trained individually.

The experiment used an ABA design. Participants practiced eight words (Set A: four words each containing /ɽɑ/ and /ɽɛ/) during the first four training sessions (Tables 1 and 2). They practiced the other eight words (Set B: four words containing /ɽi/ and /ɽo/) during the second four training sessions. This design was selected to determine whether learning generalized across post-consonantal vowel contexts. Given the small number of participants, an alternating baseline treatment design was not used. Thus, stimuli were practiced in randomized order within each training session; however, all participants followed the same blocked presentation of stimuli across the eight training sessions.

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TABLE 2. Schedule

Baseline			Training								4 wks. Post	
B1	B2	B3	T1	T2	T3	T4	T5	T6	T7	T8	P1	P2
			Set A: 8 words				Set B: 8 words					

Baseline

Three repetitions of the stimuli were recorded as probes during the baseline sessions. Audio stimuli used to elicit these productions were pre-recorded by a native speaker of Japanese (JL). The stimuli were randomized for each recording session, and presented one at a time using Microsoft Power Point software (v. 2003). A subject-controlled stimulus procedure (Logan & Pruitt, 1995) was used. Participants were seated in front of a computer monitor and produced each Japanese stimulus immediately following a pre-recorded audio model, as prompted by the program at their own pace. The participants' speech was directly recorded to the disk via a headphone microphone.

Training

Each training session lasted approximately 20 min. and was conducted by a native Japanese instructor (JL). Participants were randomly assigned to one of two training conditions: (1) conventional L2 training ("non-EMA group"), or (2) conventional L2 training with EMA-based kinematic feedback on tongue position ("EMA group").

The training procedure was similar for both groups. Participants practiced each of the speech sounds 20 times in blocked fashion, resulting in a total of 160 productions (8 words x 20 repetitions) during each training session. The instructor modeled each word written on an index card and asked participants to practice the word five times at a time. The instructor provided summarized verbal feedback regarding the articulatory positions and the timing of tongue movements (KP: knowledge of performance) after every five repetitions. Also, occasional general comments were made to keep participants motivated.

In the EMA condition, participants wore an EMA helmet (during training phases) with a small magnet (2 mm high x 2.5 mm wide x 3.7 mm long) attached approximately one cm posterior to the tongue tip using a biocompatible adhesive (Figure 1). The magnet was connected by fine wires to the analog unit. The magnet moved as the participant produced speech, and the tongue position was displayed on the main computer screen in real time and on a secondary monitor placed in front of the participant (Figures 2 and 3). Prior to each training session, a general area in the mouth was marked with a circle (e.g., the larger circle in Figure 3). The participant next produced the word "daddy," to identify the alveolar ridge region corresponding to correct production of the flap /ɾ/. The target zone for the Japanese flap was then set at a region marked slightly posterior to that area, using a mouse controlled drawing tool (e.g., the smaller circle in Figure 3). These two circles and the tongue trace are shown to the participant.

Three repetitions of 16 CVCV stimuli (Table 1) were recorded as probes at each training session, using the same recording facilities and procedures as those used for the baseline recording.

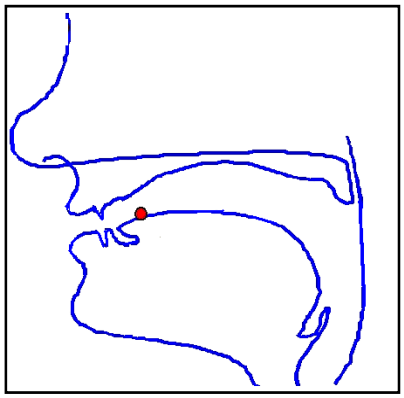

FIGURE 1. Sensor Coil Placement

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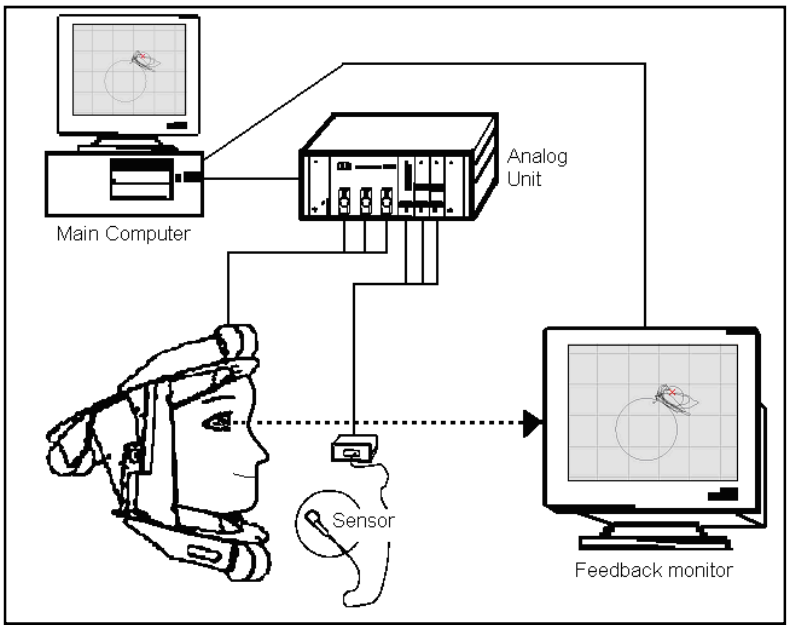

FIGURE 2. Kinematic Feedback with EMA

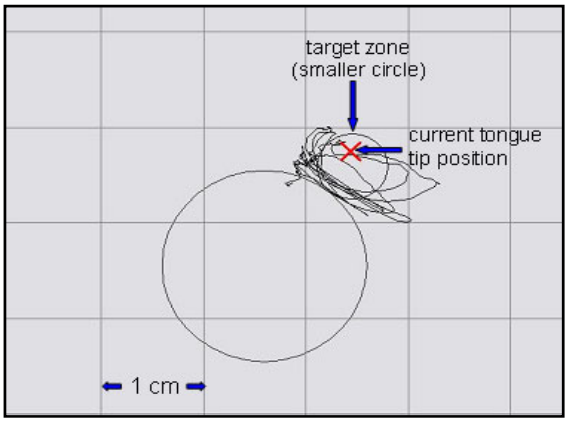

FIGURE 3. Sample EMA Screen

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Post-training/Long-term assessment

The post-training probes were recorded four weeks after the completion of the training sessions. The participants' speech was recorded using the same recording facilities and procedures as those used for baseline recording.

RESULTS

The duration of the CVCV words and flap consonants in speech produced during the baseline, training, and post-training sessions (16 words x 3 repetitions x 13 sessions x 8 participants = 4,992 recorded words) were measured from a display showing the raw waveform, spectrogram, and F_0 plot (Wavesurfer speech analysis software, v. 1.8.5). The proportion of the flap consonant relative to the disyllabic word (i.e., percent length) was calculated for analysis. This step was taken so that individual differences in the rate of speech would not affect the data analyses. Individual talker data were plotted for each phase of the experiment by word and by position of the target sound (initial or

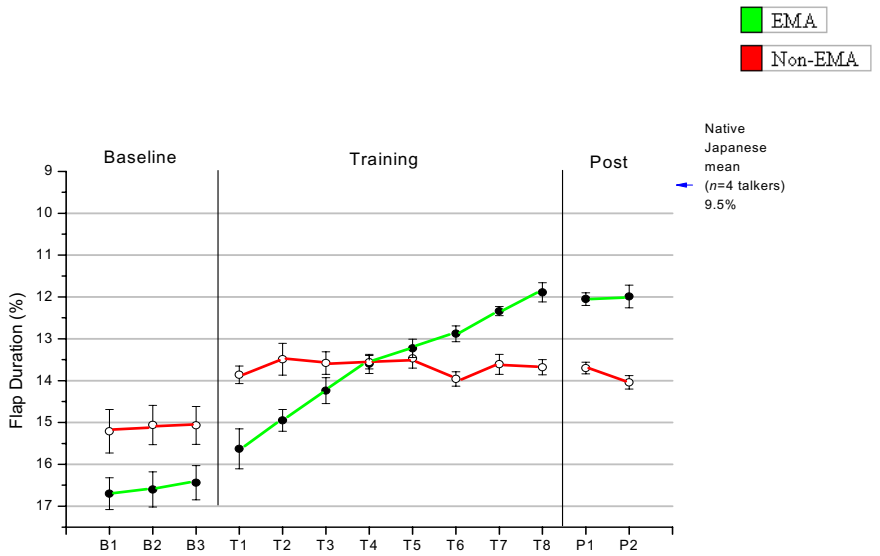
medial position). As a reliability check on duration measurement, five percent of the data (240 tokens) were randomly selected and presented to a second examiner (an ASHA certified CCC-SLP with experience in acoustic phonetics) with subject testing date and condition information removed. There was 99% agreement, with an absolute deviation of 0.5%.

As shown in Table 3, there was strong generalization between the A and B word lists. That is, during training of the A set, the duration of flaps in the B set showed almost identical progress. Therefore, the data were collapsed across word list, as shown in Figure 4.

TABLE 3. Baseline, training, and post-training data: Flap duration (% of word) of Set A and B

		B1	B2	B3	T1	T2	T3	T4	T5	T6	T7	T8	P1	P2
EMA	Set A	17	17	17	16	16	15	14	13	13	12	12	12	12
	Set B	16	16	16	15	14	14	14	13	13	12	12	12	12
Non-EMA	Set A	15	15	15	14	14	14	13	14	14	14	14	14	14
	Set B	15	15	15	14	13	14	14	13	14	14	14	14	14

Figure 4 shows the mean flap duration of probe data for the baseline, training, and post-training phases. Participants in the non-EMA condition showed 8% net improvement, with a relatively flat profile during training. Participants in the EMA condition showed 28% net improvement, with a monotonic rising pattern during training. Training effects appeared well-maintained for both the EMA and non-EMA conditions.



	EMA	Non-EMA
Baseline mean[1]	16.58	15.11
Post mean[2]	12.02	13.87
% improvement $^{(2-1)/1}$	**27.50**	**8.20**

FIGURE 4. Proportion of flap duration to whole CVCV words for two training conditions. Sessions are indicated on the x axis, and the flap consonant duration (%) is shown on the y axis. For reference, the mean value of productions by (*n*=4) native Japanese speakers are indicated by the blue arrow. Note that the y axis shows decreasing flaps duration to reflect the fact that improvement for the L2 speakers consisted of producing shorter durations that approach the Japanese norm.

Effect of vowel context

Figure 5 shows flap duration as a function of vowel context. Contrary to predictions, the results indicated no obvious vowel context effects. All vowel contexts showed stronger training effects for the EMA than the non-EMA

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conditions. The /ɽo/ data further suggest that the stronger training effects observed for the EMA condition are not solely due to low performance at baseline.

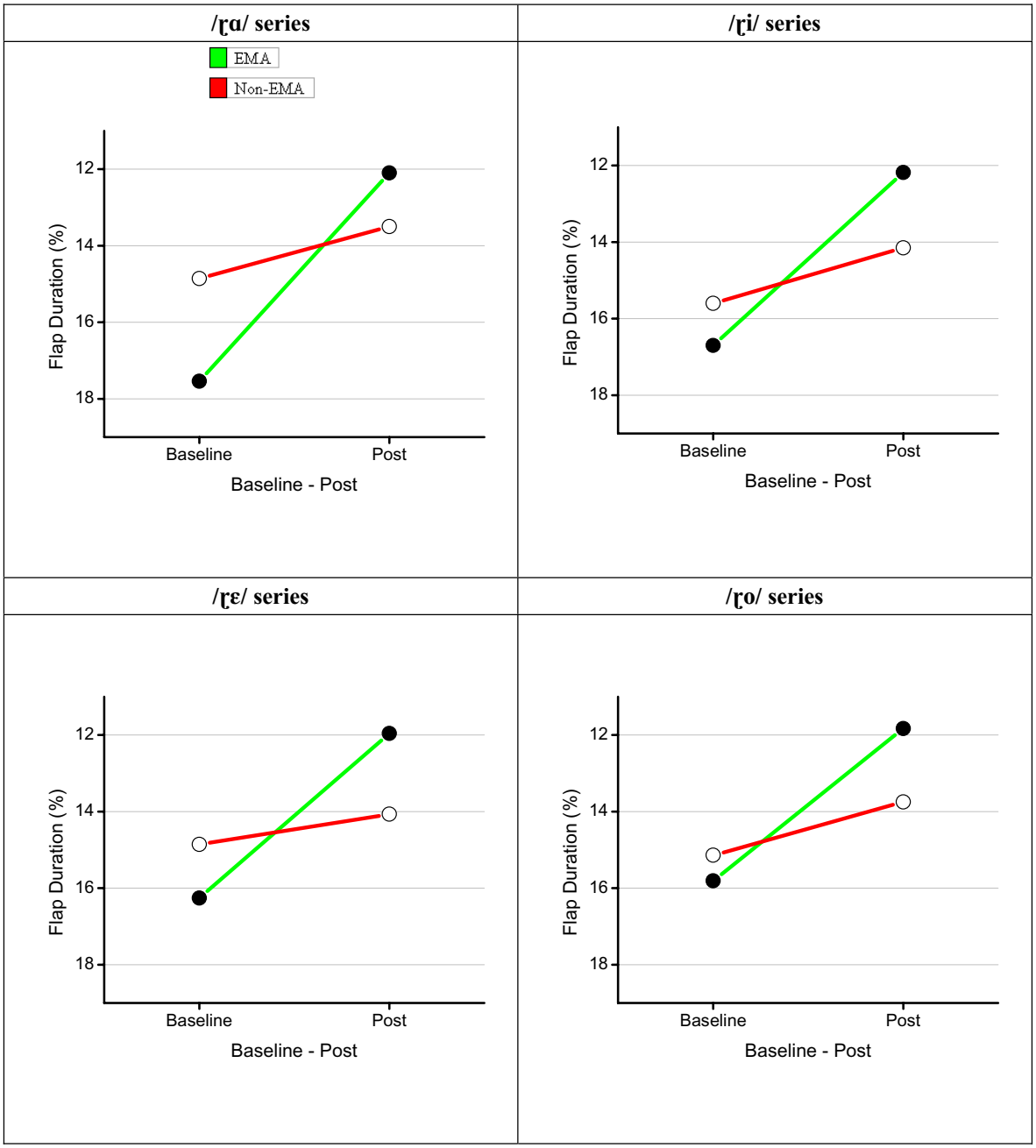

FIGURE 5. Proportion of Japanese flap duration/whole word for productions by English speakers. EMA (green) and non-EMA (red) training conditions are shown at baseline and post-training phases for different vowel contexts. Note that the y axis shows decreasing flaps duration to reflect the fact that improvement for the L2 speakers consisted of producing shorter durations that approach the Japanese norm.

Effect of familiarity

The results indicate no effect of phonological familiarity. The mean values of familiar and unfamiliar learning conditions were almost completely identical, with no statistically significant differences. Thus, contrary to the

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predictions of both theories of L2 acquisition investigated (e.g. phonotactic familiarity view and SLM), neither condition produced better learning effects than the other.

Effect size analysis

To further assess the improvement of flap consonant production between the EMA and non-EMA conditions, Cohen's *d* analyses were conducted. Cohen's *d* values represent an effect size that measures the magnitude of a treatment or training effect (Beeson & Robey, 2006). Unlike significance tests, these figures are independent of sample size. Thus, this method is commonly used to examine the degree of difference between baseline and post-training phases (Lipsey & Wilson, 1993). Cohen's *d* values were first calculated for each group of 16 words, yielding a unique *d* value for each participant. The mean Cohen's *d*, or *d*-prime, was then computed per training condition (EMA vs. Non-EMA) for each individual. The *d*-prime for the individuals in the EMA group were uniformly greater than those in the non-EMA group with average scores of 13.91 and 4.78, respectively (see Table 4).

TABLE 4. Effect size: Cohen's *d* analysis

Condition	Participant	Cohen's *d*	*d*-Prime
EMA	1	12.13	13.91
	2	12.38	
	3	13.98	
	4	17.14	
Non-EMA	5	2.71	4.78
	6	7.60	
	7	4.39	
	8	4.18	

Cohen (1988) provided default benchmarks to evaluate *d* scores. A small effect size is 0.2., a medium effect size is 0.5, and a large effect size is 0.8. A small effect size means the upper half of the post-training data exceeds 57.9% of the baseline data, a medium effect size means that the upper half of the post-training data exceeds 69.1% of the baseline data, and a large effect size means that the upper half of the post-training data exceeds the lower 78.8% of the baseline data (Cohen, 1988). Cohen also strongly recommends comparison with actual *d* values in the associated literature to gain a better idea of the effectiveness of a given intervention. Two recent, closely related word training studies have reported effects sizes ranging between 0.02 and 1.008 (Verdugo & Belmonte, 2007) and 0.06 and 3.05 (Rott, 2007).

In summary, considering both Cohen's benchmark guidelines and *d*-values of other L2 training studies, the effect sizes in the current study suggest positive training effects for both conditions. The current data also indicate a substantially greater training effect in the EMA condition than in the non-EMA condition.

DISCUSSION

This study was designed to address several issues: (1) the relationship between limb and speech motor learning, (2) hypothesized positive EMA training effects for healthy individuals, and (3) predictions concerning the strength of training as a function of vowel context and familiarity (initial versus medial word position).

Eight normal adult native speakers of American English participated in speech training to learn the Japanese post-alveolar flap consonant /ɽ/. Each participant attended three baseline recording, eight training, and two post-training recording sessions either in EMA or non-EMA condition. The duration of the each probe was measured, and relative duration (%) of the flap consonant in each disyllabic word was computed.

The results suggest EMA-based visual biofeedback concerning tongue-tip position facilitated healthy adults' learning of novel speech sounds, as determined by visual inspection of acquisition/generalization/maintenance patterns and a greater *d*-prime effect size (EMA condition: 13.91; non-EMA condition: 4.78). Measurement of flap duration, a known correlate of the flap vs. stop distinction (Kent & Read, 2002; Shah, 2004), indicated progressively more appropriate flap durations in EMA condition, with less marked improvement in the non-EMA condition (see Figure 4). Although it remains important to conduct a formal perceptual study, informal perceptual judgment by the investigator suggests that the participants improved and sounded closer to native Japanese productions.

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According to Swinnen (1996), augmented feedback is most beneficial for retention when utilized with intrinsic feedback, but not as a substitute for intrinsic feedback. In such conditions, the degree of compatibility among action patterns performed simultaneously and the availability of intrinsic feedback are critical factors. In the current study, it may be the case that (1) participants used the augmented visual feedback information provided during training to update and refine the processing of the intrinsic (sensory) information feedback sources, and (2) this allowed an effective visual and/or kinesthetic guidance of movement as practice continued. The learned motor behavior was retained through the post-training recording phase conducted four weeks after the last training. Taken together, the overall result suggest similar effects of augmented visual feedback for speech learning (Dowd, Smith, & Wolfe, 1998), speech remediation for disordered individuals (Katz et al., 1999, 2002), and limb motor learning (De Nil & Lafaille, 2002; Swinnen et al., 1997).

With respect to vowel context predictions, there were no substantial differences across the Japanese disyllables /ɽɑ/, /ɽɛ/, /ɽi/, or /ɽo/. All the words were produced with essentially the same level of proficiency across different post-consonantal vowels. Although it is potentially problematic to argue from a lack of statistical differences across conditions, this pattern would nevertheless provide qualified support for the view that the Japanese flap was acquired as one GMP, with different parameters filled in across vowel contexts (Schmidt, 1975).

The observed lack of positional familiarity effects (word-initial vs. –medial) did not match the predictions of either the phonotactic familiarity view (Davidson, 2005) or the Speech Learning Model (SLM) (Flege, 1995). One possible explanation for the present results is that participants' learning of Japanese flap was influenced by an interaction between these properties. Alternatively, the task of learning Japanese disyllabic words may have placed other demands on the participants (such as producing the more monophthongal vowels or slightly different stress/mora patterns of Japanese) that masked any fine-grained differences such as syllable position. Clearly, additional research will be necessary to determine the role of word position (and associated phoneme vs. allophone factors) in L2 learning.

These familiarity issues relate to other recent studies of flap acquisition by native speakers of American English. For instance, Pruitt and colleagues noted that Japanese learners were better than American learners at acquiring the Hindi dental and retroflex stop consonant contrast in CV syllables (Pruitt, Jenkins, & Strange, 2006). Japanese has a phonemic distinction between dental stop and alveolar flap. The findings were taken as evidence that allophonic experience with stops and flaps (e.g., in American English) may be detrimental to perception of the new contrast. Given the current unexpected findings concerning syllable position and familiarity (due to allophonic variation), the present results suggest L2 learning studies could benefit from including both CV and CVCV target stimuli.

To summarize, the results of this study suggest that augmented feedback provided by an EMA system can facilitate the normal adult speakers' learning of novel speech sounds. However, this conclusion must be weighed with caution for several reasons. First, the experimental design was not double-blinded. Although the instructor (JL) followed a written procedure designed to guide both groups (across the two training conditions) in the same manner, instructor bias could have affected results. This type of methodological shortcoming is unfortunately rather common with most of the speech training/remediation studies to date (Ballard, Maas, & Robin, 2007; Grillo, 1997; Hardison, 1998; Kim, 2007). Therefore, this experiment must be duplicated either with a double-blinded instructor design or via automated instruction with no possibility of instructor bias, before the present results can be considered valid. Second, this study used an ABA design. Thus, it did not explore skill transfer and generalization, as could be examined with a multiple baseline design. Also, because the training was ended after a fixed number of sessions (i.e. after 8 sessions), participants were not trained to mastery criteria. With further trainings, the speech production of the participants in EMA condition may have reached the native Japanese speakers' norm.

CONCLUSIONS

The results of this unblinded study using an ABA design suggest similar principles of motor learning theory apply to limb and speech movements. Concurrent visual feedback produced a positive effect in speech learning, suggesting that augmented visual feedback could strengthen intrinsic feedback processing mechanisms. The present data are based on one acoustical feature, i.e., flap duration. To bolster these data, investigations are currently in progress examining a cue in the spectral domain (i.e., flanking vowel formant frequency transitions). It is also important to establish perceptual measurements of the obtained data. Future investigations are planned to obtain identification/rating data from native speakers of Japanese. If the present positive training outcomes are confirmed in additional analyses, they will add to a growing body of research supporting the positive effect of augmented visual feedback on speech learning.

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